

SkillClaw

Collective Skill Evolution for Multi-User LLM Agent Ecosystems

arxiv.org/abs/2604.08377

Let Skills Evolve Collectively with Agentic Evolver

Agent Skills Remain Static After Deployment

Core Problem

Users manually install skills from a centralized hub. Solutions discovered during interaction rarely persist beyond individual sessions. As similar tasks recur, the same patterns of failure are repeatedly observed — but the system does not improve.

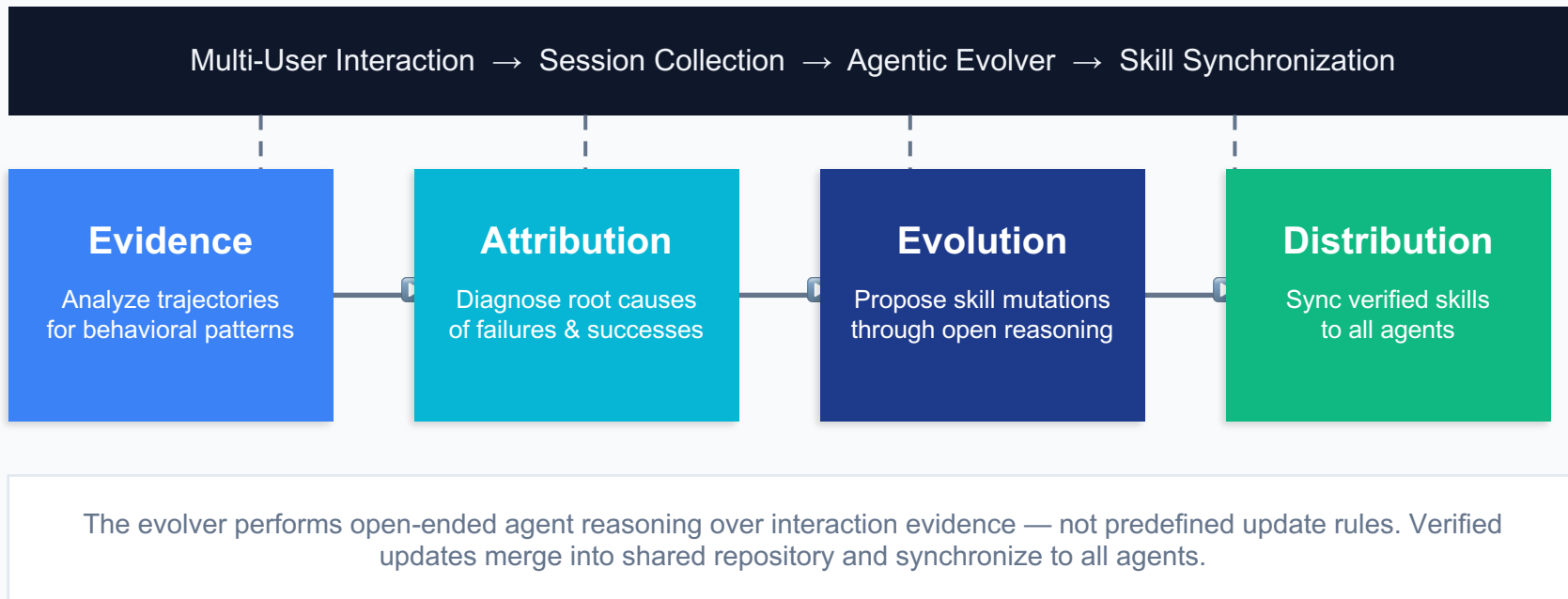
Static Paradigm

- Skills installed once, never updated
- Failures rediscovered per user
- No knowledge accumulation
- Memory tied to specific instances

SkillClaw Paradigm

- Continuous skill evolution from usage
- Failures become shared improvements
- System-level knowledge accumulation
- Cross-user trajectory aggregation

Closed-Loop Pipeline: From Multi-User Interaction to Shared Skill Evolution



Compatible Systems:

OpenClaw • CoPaw • IronClaw • PicoClaw • ZeroClaw • NanoClaw • NemoClaw

Agentic Evolver: Evidence → Attribution → Evolution in Three Stages

01

Evidence

- Analyze trajectories from multi-user sessions
- Identify behavioral patterns & success signals
- Group evidence by skill under diverse contexts

02

Attribution

- Diagnose root causes of failures
- Distinguish success patterns from noise
- Identify conditions where skill breaks down

03

Evolution

- Propose skill mutations via open reasoning
- Refine existing skills or create new ones
- Submit candidates to validation pipeline

Cross-User Trajectory Aggregation Exposes Skill Behavioral Boundaries

Why Cross-User Aggregation?

- Diverse contexts reveal complementary signals
- Single user rarely generates enough signal to separate generalizable improvements from idiosyncratic fixes
- Same skill under different users exposes both working conditions and failure modes
- Trajectories preserve full action–feedback causal chains

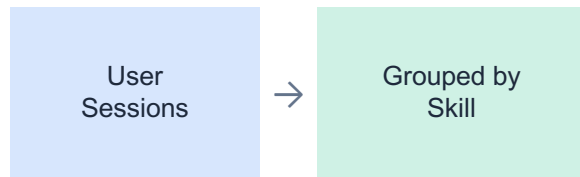
Key Insight

A single user exercising a skill under consistent contexts cannot generate the complementary views needed to identify true behavioral boundaries.

Compatible Claw-Style Systems



Evidence Aggregation



Case Study: Multimodal Creative Pipeline — 6 Days of Evolution

1 of 6
Accepted

Day	Candidate Skill	Outcome	Rationale
1	validate-tmp-workspace-inputs	Accept	Upgrade to new best pool
2	multimodal-input-validation-and-setup	Reject	Keep current best pool
3	multimodal-creative-task-pipeline	Reject	Keep current best pool
4	multimodal-creative-task-pipeline (impr.)	Reject	Keep current best pool
5	pipeline + validate-required-input-files	Reject	Keep current best pool
6	multimodal-creative-task-pipeline (cand.)	Reject	Not admitted to next cycle

Interpretation: Conservative verification prevents regressions. Only Day 1 candidate passed validation — subsequent candidates were rejected to maintain skill pool integrity.

Case Study: Git Push Auth-Fallback Skill — Iterative Refinement Over 6 Days

3 of 6

Accepted

Day	Candidate Skill	Outcome	Rationale
1	git-push-with-auth-fallback	Accept	Add to Safety best pool
2	git-push-with-auth-fallback	Accept	Unified patch/bundle filenames
3	git-push + git-clone-to-directory	Accept	Auth-alternative paths & audit
4	(none — same pool retest)	Accept	Validator confirmed best pool
5	git-push-with-auth-fallback	Reject	No improvement — keep best pool
6	git-push-with-auth-fallback	Reject	No improvement — plateau reached

Interpretation: Iterative refinement succeeded — Days 1–3 yielded accepted improvements before the skill reached a natural plateau on Days 5–6.

Key Takeaways

Three Properties That Distinguish SkillClaw

1

Collective Evolution

Individual interactions contribute to a shared, continuously improving skill ecosystem across all users.

2

Fully Automatic

Skill evolution is driven by runtime interaction without manual curation or explicit user intervention.

3

Agentic Evolution Paradigm

Skill updates are produced through open-ended reasoning rather than predefined update rules.

Implications for Continuously Improving Agent Systems

SkillClaw enables LLM agent ecosystems to transform runtime failures into cumulative improvements — shifting from static skill libraries to self-improving, collectively-evolving systems.

Evaluated on WildClawBench with Qwen3-Max backbone